



TRIOS CYBER

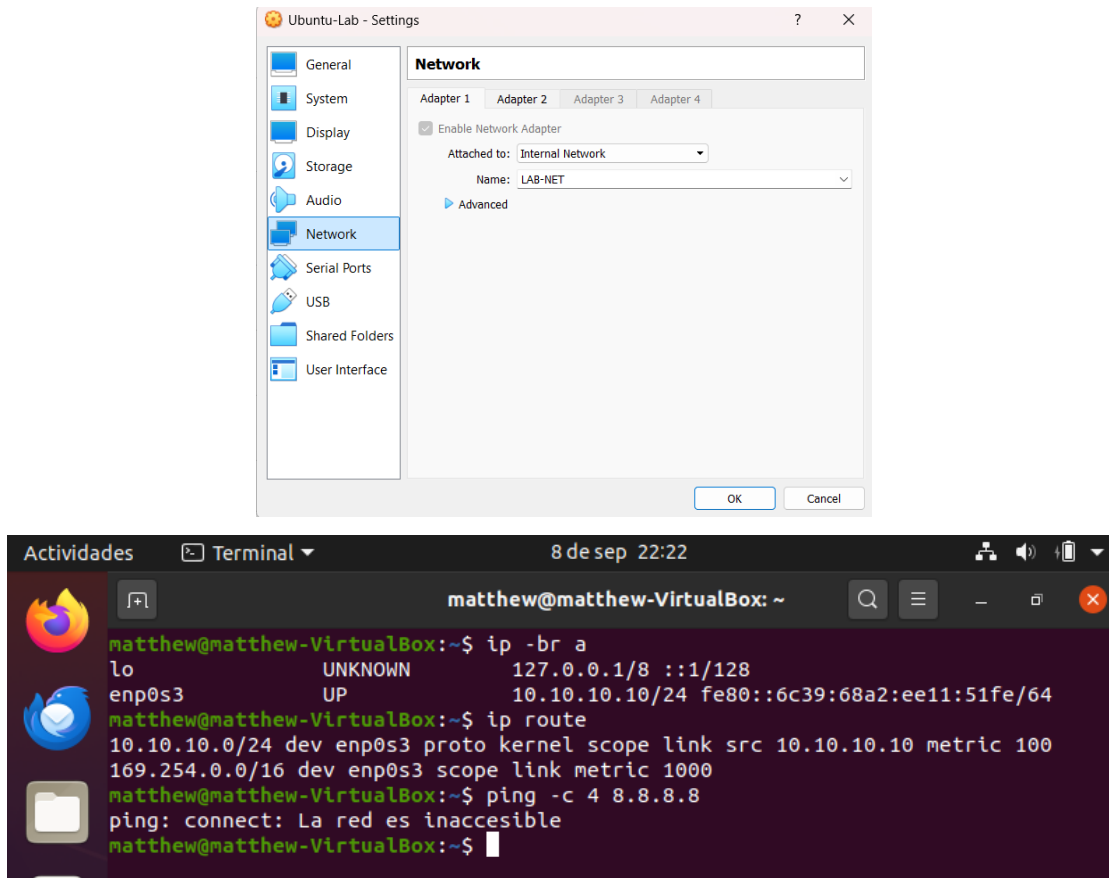
LABORATORY PRACTICE 7

Simulated Authentication Scenario and Indicator Collection

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1. Isolated Laboratory Environment Verification

The Ubuntu laboratory endpoint was placed on the Oracle VirtualBox Internal Network named LAB-NET. The active interface enp0s3 used the isolated address 10.10.10.10/24, and an external connectivity test to 8.8.8.8 failed. This confirmed that the simulated activity was performed inside the controlled laboratory network rather than against an external system.



2. Initial Indicator Collection

Before generating the authentication events, the endpoint identity and time were recorded from Ubuntu. The active account was matthew, the hostname was matthew-VirtualBox, the system time was Sep 08 2026 at approximately 22:23:10 (-05), and the endpoint address was 10.10.10.10/24. These values established reference indicators for correlating the later log entries.

```
matthew@matthew-VirtualBox:~$ whoami
matthew
matthew@matthew-VirtualBox:~$ hostname
matthew-VirtualBox
matthew@matthew-VirtualBox:~$ date
mar 08 sep 2026 22:23:10 -05
matthew@matthew-VirtualBox:~$ ip -br a
lo                UNKNOWN      127.0.0.1/8  ::1/128
enp0s3            UP          10.10.10.10/24  fe80::6c39:68a2:ee11:51fe/64
matthew@matthew-VirtualBox:~$
```

3. Safe Local Authentication Simulation

A harmless local scenario was first generated by attempting to switch to the matthew account three times with intentionally incorrect passwords. Each `su - matthew` attempt returned an authentication failure. The number of attempts was kept low and the activity remained on the authorized Ubuntu endpoint.

```

matthew@matthew-VirtualBox:~$ su - matthew
Contraseña:
su: Fallo de autenticación
matthew@matthew-VirtualBox:~$ su - matthew
Contraseña:
su: Fallo de autenticación
matthew@matthew-VirtualBox:~$ su - matthew
Contraseña:
su: Fallo de autenticación
matthew@matthew-VirtualBox:~$

```

4. Local Authentication Event Review

The operating-system journal was searched with journalctl and filters for authentication failure, FAILED SU, and pam_unix. The resulting records showed the three controlled failures, including the matthew account, the matthew-VirtualBox host, the su process, PAM authentication context, pts/0 terminal information, and event timestamps. A broader su-related review was also used to preserve surrounding session context.

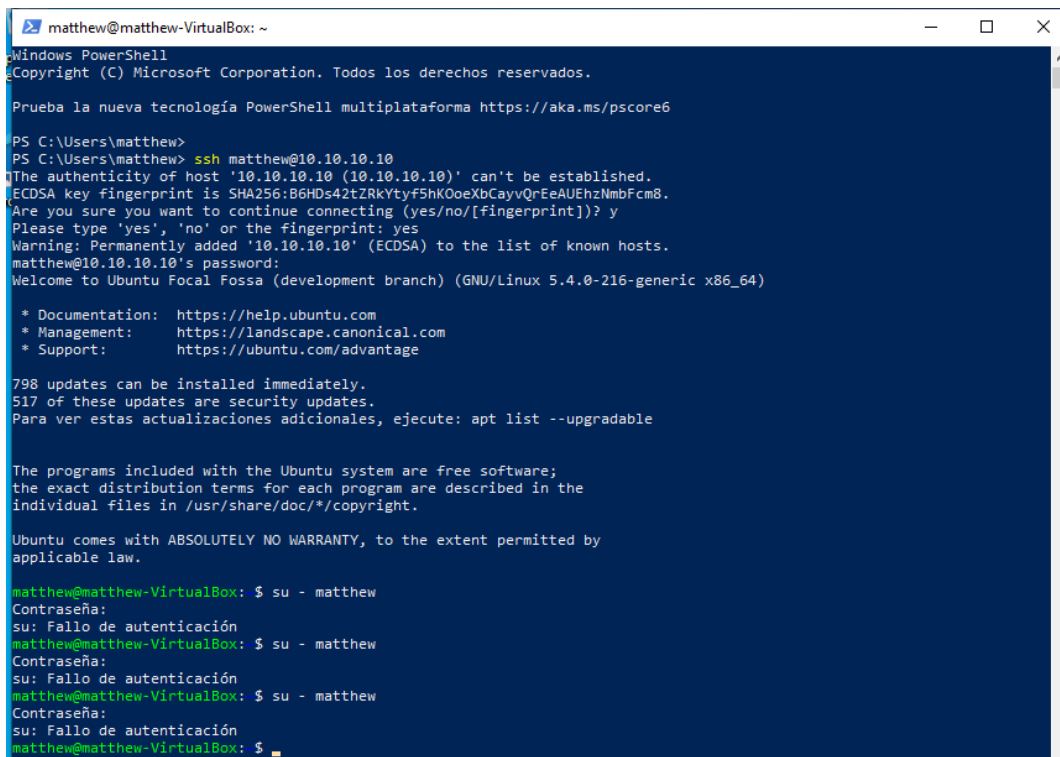
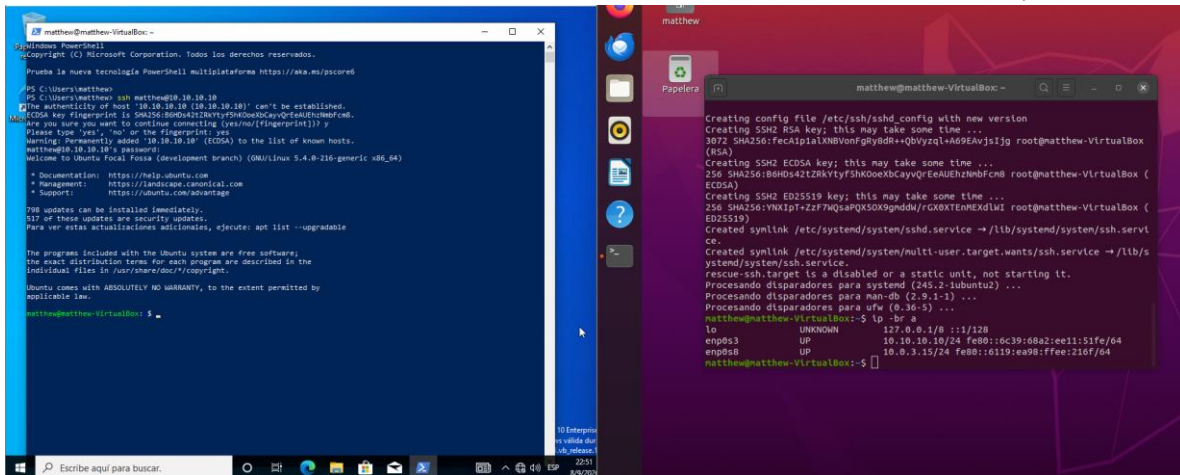
```

matthew@matthew-VirtualBox:~$ sudo journalctl --since "10 minutes ago" | grep -Ei "authentication failure|FAILED SU|pam_unix"
[sudo] contraseña para matthew:
sep 08 22:20:56 matthew-VirtualBox gdm-launch-environment[666]: pam_unix(gdm-launch-environment:session): session opened for user gdm by (uid=0)
sep 08 22:20:56 matthew-VirtualBox systemd[762]: pam_unix(systemd-user:session): session opened for user gdm by (uid=0)
sep 08 22:21:17 matthew-VirtualBox gdm-password[1093]: pam_unix(gdm-password:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:21:20 matthew-VirtualBox gdm-password[1093]: pam_unix(gdm-password:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:21:20 matthew-VirtualBox gdm-password[1093]: pam_unix(gdm-password:session): session opened for user matthew by (uid=0)
sep 08 22:21:20 matthew-VirtualBox systemd[1108]: pam_unix(systemd-user:session): session opened for user matthew by (uid=0)
sep 08 22:21:29 matthew-VirtualBox gdm-launch-environment[666]: pam_unix(gdm-launch-environment:session): session closed for user gdm
sep 08 22:22:38 matthew-VirtualBox pkexec[2176]: pam_unix(polkit-1:session): session opened for user root by (uid=1000)
sep 08 22:24:09 matthew-VirtualBox su[2206]: pam_unix(su-l:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:24:12 matthew-VirtualBox su[2206]: pam_unix(su-l:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:24:12 matthew-VirtualBox su[2206]: pam_unix(su-l:auth): authentication failure; logname= uid=1000 euid=0 tty=pts/0 ruser=matthew rhost= use
r=matthew
sep 08 22:24:13 matthew-VirtualBox su[2206]: FAILED SU (to matthew) matthew on pts/0
sep 08 22:24:16 matthew-VirtualBox su[2207]: pam_unix(su-l:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:24:17 matthew-VirtualBox su[2207]: pam_unix(su-l:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:24:17 matthew-VirtualBox su[2207]: pam_unix(su-l:auth): authentication failure; logname= uid=1000 euid=0 tty=pts/0 ruser=matthew rhost= use
r=matthew
sep 08 22:24:18 matthew-VirtualBox su[2207]: FAILED SU (to matthew) matthew on pts/0
sep 08 22:24:22 matthew-VirtualBox su[2208]: pam_unix(su-l:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:24:23 matthew-VirtualBox su[2208]: pam_unix(su-l:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:24:23 matthew-VirtualBox su[2208]: pam_unix(su-l:auth): authentication failure; logname= uid=1000 euid=0 tty=pts/0 ruser=matthew rhost= use
r=matthew
sep 08 22:24:25 matthew-VirtualBox su[2208]: FAILED SU (to matthew) matthew on pts/0
sep 08 22:29:19 matthew-VirtualBox sudo[2243]: pam_unix(sudo:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:29:21 matthew-VirtualBox sudo[2243]: pam_unix(sudo:auth): Couldn't open /etc/securetty: No existe el archivo o el directorio
sep 08 22:29:21 matthew-VirtualBox sudo[2243]: matthew : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/sudo journalctl --since 10 minu
tes ago
sep 08 22:29:21 matthew-VirtualBox sudo[2243]: pam_unix(sudo:session): session opened for user root by (uid=0)
sep 08 22:29:21 matthew-VirtualBox sudo[2245]: root : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/journalctl --since 10 minutes a
go
sep 08 22:29:21 matthew-VirtualBox sudo[2245]: pam_unix(sudo:session): session opened for user root by (uid=0)
sep 08 22:29:21 matthew-VirtualBox sudo[2245]: pam_unix(sudo:session): session closed for user root
sep 08 22:29:21 matthew-VirtualBox sudo[2243]: pam_unix(sudo:session): session closed for user root
sep 08 22:32:49 matthew-VirtualBox systemd[1]: anacron.service: Succeeded.
sep 08 22:35:21 matthew-VirtualBox systemd[1]: systemd-tmpfiles-clean.service: Succeeded.
sep 08 22:36:23 matthew-VirtualBox /usr/lib/gdm3/gdm-x-session[1124]: (EE) No surface to present from.
sep 08 22:36:23 matthew-VirtualBox /usr/lib/gdm3/gdm-x-session[1124]: (EE) No surface to present from.
sep 08 22:36:28 matthew-VirtualBox systemd[1108]: gnome-launched-gnome-control-center.desktop-2310.scope: Succeeded.
sep 08 22:37:31 matthew-VirtualBox sudo[2332]: matthew : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/sudo journalctl --since 10 minu
tes ago
sep 08 22:37:31 matthew-VirtualBox sudo[2334]: root : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/journalctl --since 10 minutes a
go
sep 08 22:37:31 matthew-VirtualBox sudo[2334]: pam_unix(sudo:session): session opened for user root by (uid=0)

```

5. Remote Authentication Scenario Preparation

To obtain a network source indicator, the Windows and Ubuntu laboratory machines were used together on the isolated LAB-NET. Ubuntu provided an SSH service on TCP port 22, while the Windows laboratory endpoint used 10.10.10.20 to reach Ubuntu at 10.10.10.10. A controlled successful connection was first used to confirm that the two authorized lab endpoints could communicate before generating failed remote logins.



6. SSH Service and Remote Source Verification

The Ubuntu journal confirmed that sshd was listening on port 22 and recorded a successful SSH session for matthew from source address 10.10.10.20. This provided a clear network relationship between the Windows source endpoint and the Ubuntu target and established the source IP that would later appear in the failed-authentication evidence.

```

matthew@matthew-VirtualBox:~$ sudo journalctl --since "10 minutes ago" | grep -Ei "ssh|Failed password|authentication failure"
sep 08 22:44:55 matthew-VirtualBox systemd[1211]: Starting Start gnome-keyring as SSH agent...
sep 08 22:44:55 matthew-VirtualBox systemd[1211]: Finished Start gnome-keyring as SSH agent...
sep 08 22:44:55 matthew-VirtualBox systemd[1211]: Started OpenSSH Agent.
sep 08 22:44:55 matthew-VirtualBox systemd[1211]: ssh-agent.service: Succeeded.
sep 08 22:44:56 matthew-VirtualBox gnome-keyring-ssh.desktop[1503]: SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
sep 08 22:45:02 matthew-VirtualBox systemd[747]: Stopping Start gnome-keyring as SSH agent...
sep 08 22:45:02 matthew-VirtualBox systemd[747]: gnome-keyring-ssh.service: Succeeded.
sep 08 22:45:02 matthew-VirtualBox systemd[747]: Stopped Start gnome-keyring as SSH agent.
sep 08 22:45:12 matthew-VirtualBox systemd[747]: gpg-agent-ssh.socket: Succeeded.
sep 08 22:45:12 matthew-VirtualBox systemd[747]: Closed GnuPG cryptographic agent (ssh-agent emulation).
sep 08 22:46:06 matthew-VirtualBox sudo[2738]: matthew : TTY=pts/0 ; PWD=/home/matthew ; USER=root ; COMMAND=/usr/bin/apt install openssh-server -y
sep 08 22:46:19 matthew-VirtualBox useradd[3159]: new user: name=sshd, UID=123, GID=65534, home=/run/sshd, shell=/usr/sbin/nologin, from=none
sep 08 22:46:19 matthew-VirtualBox usermod[3167]: change user 'sshd' password
sep 08 22:46:19 matthew-VirtualBox chage[3174]: changed password expiry for sshd
sep 08 22:46:21 matthew-VirtualBox sshd[3322]: Server listening on 0.0.0.0 port 22.
sep 08 22:46:21 matthew-VirtualBox sshd[3322]: Server listening on :: port 22.
sep 08 22:51:48 matthew-VirtualBox sshd[5068]: Accepted password for matthew from 10.10.10.20 port 49673 ssh2
sep 08 22:51:48 matthew-VirtualBox sshd[5068]: pam_unix(sshd:session): session opened for user matthew by (uid=0)
sep 08 22:52:32 matthew-VirtualBox su[5225]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=pts/1 ruser=matthew rhost=
t= user=matthew
sep 08 22:52:40 matthew-VirtualBox su[5226]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=pts/1 ruser=matthew rhost=
t= user=matthew
sep 08 22:52:51 matthew-VirtualBox su[5228]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=pts/1 ruser=matthew rhost=
t= user=matthew

```

7. Controlled Remote Failed Authentication Attempts

From Windows PowerShell, an SSH login to matthew@10.10.10.10 was attempted with an incorrect password. The client displayed repeated Permission denied messages and terminated the authentication attempt without gaining access. The test remained limited to the authorized account and isolated laboratory network.

```

PS C:\Users\matthew> ssh matthew@10.10.10.10
matthew@10.10.10.10's password:
Permission denied, please try again.
matthew@10.10.10.10's password:
Permission denied, please try again.
matthew@10.10.10.10's password:
matthew@10.10.10.10: Permission denied (publickey,password).

```

8. Remote Authentication Log Analysis

The Ubuntu journal was then filtered for ssh, Failed password, and authentication failure events. The log correlated the remote test with the matthew account and source address 10.10.10.20. Three Failed password entries were recorded at 22:57:31, 22:57:36, and 22:57:41 using remote source port 49674 and SSH version 2. The same sequence also contained pam_unix(sshd:auth) authentication failure data, the sshd process identifier, and the final connection-reset message.

```

matthew@matthew-VirtualBox:~$ sudo journalctl --since "10 minutes ago" | grep -Ei "ssh|Failed password|authentication failure"
sep 08 22:51:48 matthew-VirtualBox sshd[5068]: Accepted password for matthew from 10.10.10.20 port 49673 ssh2
sep 08 22:51:48 matthew-VirtualBox sshd[5068]: pam_unix(sshd:session): session opened for user matthew by (uid=0)
sep 08 22:52:32 matthew-VirtualBox su[5225]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=pts/1 ruser=matthew rhost=
t= user=matthew
sep 08 22:52:40 matthew-VirtualBox su[5226]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=pts/1 ruser=matthew rhost=
t= user=matthew
sep 08 22:52:51 matthew-VirtualBox su[5228]: pam_unix(su-l:auth): authentication failure; logname=matthew uid=1000 euid=0 tty=pts/1 ruser=matthew rhost=
t= user=matthew
sep 08 22:56:58 matthew-VirtualBox sshd[5181]: Received disconnect from 10.10.10.20 port 49673:11: disconnected by user
sep 08 22:56:58 matthew-VirtualBox sshd[5181]: Disconnected from user matthew 10.10.10.20 port 49673
sep 08 22:56:58 matthew-VirtualBox sshd[5068]: pam_unix(sshd:session): session closed for user matthew
sep 08 22:57:29 matthew-VirtualBox sshd[5392]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.20 user=matthew
sep 08 22:57:31 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:36 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: Failed password for matthew from 10.10.10.20 port 49674 ssh2
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: Connection reset by authenticating user matthew 10.10.10.20 port 49674 [preauth]
sep 08 22:57:41 matthew-VirtualBox sshd[5392]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.10.10.20 user=matthew

```

9. Observable Indicator Summary

The local and remote scenarios produced a set of observable indicators that can be used during a basic investigation. The remote SSH test was particularly useful because the operating-system logs preserved both host-based authentication information and a network source address.

Indicator	Observed Value	Investigation Relevance
Laboratory environment	VirtualBox Internal Network: LAB-NET	Confirms the activity remained isolated
Username	matthew	Account involved in the simulated authentication activity
Hostname	matthew-VirtualBox	Ubuntu endpoint that recorded the events
Target / endpoint IP	10.10.10.10/24	Ubuntu address inside LAB-NET

Indicator	Observed Value	Investigation Relevance
Source IP	10.10.10.20	Windows lab endpoint recorded by sshd
Timestamp	Sep 08 2026; remote failures 22:57:31-22:57:41	Correlates the test with journal events
Process / service	sshd; pam_unix(sshd:auth)	Processes/components reporting remote authentication
Remote source port	49674	Client-side port recorded in failed SSH events
Destination service	SSH / TCP 22	Authorized local service used for the remote scenario
Event result	Failed password / authentication failure	Indicates unsuccessful authentication
Attempt count	3 remote failed passwords	Safe limited simulation without account lockout

10. Event Correlation Review

The evidence demonstrates how a simulated authentication event can be reconstructed from several observable fields. The initial commands established the expected user, host, time, and endpoint address; journalctl then preserved local PAM failures; and the remote SSH activity added the Windows source IP, SSH service, client port, process identifier, and exact failure timestamps. Together, these indicators provide a concise event timeline without requiring interaction with systems outside the lab.

Step	Activity	Key Evidence	Result
1	Initial baseline	matthew / matthew-VirtualBox / 10.10.10.10	Reference state recorded
2	Local failed logins	su - matthew	Three authentication failures generated
3	Local log review	pam_unix / FAILED SU	Local event details correlated
4	Remote connectivity	10.10.10.20 -> 10.10.10.10:22	Authorized SSH path verified
5	Remote failed logins	Windows SSH client	Three incorrect password attempts
6	Remote log review	sshd / pam_unix / source 10.10.10.20	Network and host indicators collected

11. Conclusions

The harmless simulated authentication scenario was successfully completed inside the isolated LAB-NET environment. The Ubuntu endpoint recorded controlled local and remote authentication failures, while the collected evidence identified the username, hostname, timestamps, endpoint address, remote source IP, SSH service, process and PAM context, and relevant connection details. The remote SSH failures from 10.10.10.20 demonstrated how operating-system logs can preserve both endpoint and network indicators that can later be correlated during an alert investigation.